Covering 1024 syndromes with 50 columns

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Abstract

We exhibit a binary linear $[50, 40]_2$ code of covering radius 2, so $\ell_2(10, 2) \leq 50$, one column below the Kaikkonen–Rosendahl length 51 that has stood since 2003 and that still seeds the $R = 2$ family of Davydov–Marcugini–Pambianco (arXiv:2511.02542). The new matrix admits a $(2, 0)$-partition into ten blocks, so Construction QM_2^2 propagates it to exhaustively verified codes of lengths 815 and 1631 at $r = 18$ and $r = 20$, and to the family $n = 51 \cdot 2^{r/2-5} - 1$ of asymptotic density 2601/2048. The matrices, verifiers, and LaTeX source are at `https://github.com/wustep/maths`. The pin is `problems/covering/share/2026-08-24/` at commit `82b5b85bd14bfa42ac7439adc7c7c956c3f84eeb`. Every number is re-checked from the files in that pin.

Computation

The author orchestrated large-language-model agents through Grok Bot. GPT-5.6 Sol found the 50-column matrix by simulated annealing on column sets in $\mathbb{F}_2^{10}$, starting from the Kaikkonen–Rosendahl 51-column set of 2003. Claude Opus 5 found the 10-block $(2, 0)$-partition of that matrix, implemented Construction QM_2^2, and wrote two independent verifiers with opposite algorithms (Python and Rust) that recheck the covering condition and the partition from the committed matrix files.

1 Definitions and background

1.1 Covering radius

Let $C \leq \mathbb{F}_2^n$ be a binary linear code of codimension r, so $|C| = 2^{n-r}$. The *covering radius* of C is the least R such that every vector of $\mathbb{F}_2^n$ lies within Hamming distance R of some codeword. Equivalently, if $\boldsymbol{H}$ is an $r \times n$ parity-check matrix of C with column set $S \subset \mathbb{F}_2^r$, the covering radius is the least R such that every syndrome is a sum of at most R columns of $\boldsymbol{H}$. Write $\ell_2(r, R)$ for the least length of a binary linear code with codimension r and covering radius R.

A binary linear code of length n and dimension $k = n - r$ is an $[n, k]_2$ code. An $[n, k]_2$ code of covering radius R is written $[n, k]_2\, R$. For $R = 2$ the covering condition on the column set is

$$\{0\} \cup S \cup (S + S) = \mathbb{F}_2^r, \qquad S + S = \{\, s + s' : s, s' \in S \,\}. \tag{1}$$

A $(2, 0)$-*partition* of the columns (Definition 3.2 of [1]) is a partition into nonempty blocks such that every syndrome, including 0, is a sum of at most two columns from *distinct* blocks. Write $p(\boldsymbol{H})$ for the number of blocks of such a partition.

1.2 Green's Problem 40

The covering density of an $[n, n - r]_2\, 2$ code is the exact rational

$$\mu = \frac{1 + n + \binom{n}{2}}{2^r},$$

the number of radius-2 representations divided by the number of syndromes. When (1) holds, $\mu \geq 1$. Green's Problem 40 [4] concerns the asymptotic quantity

$$f(2) = \liminf_{r\to\infty} \mu\bigl(\ell_2(r,2), r\bigr).$$

For an infinite family of $[n(r), n(r)-r]_2\,2$ codes, write $\bar{\mu}(2)$ for the $\liminf$ of μ along the family. The optimal lengths satisfy $\ell_2(r,2) \leq n(r)$, so any family bound $\bar{\mu}(2) \leq c$ implies $f(2) \leq c$. Whether $f(2) = 1$ is open.

The bound $f(2) \leq 729/512 \approx 1.42383$ held from 1992 to 2025. Davydov, Marcugini, and Pambianco [1] improved it to $2704/2048 \approx 1.32031$ in November 2025. Their tool is a construction, QM_2^2, that produces an infinite family from one short radius-2 code. The construction applies when the seed admits a $(2,0)$-partition into few enough blocks. Section 2.5 states its exact input and output.

1.3 The seed at $r = 10$

Table 5.1 of [1], row $r = 10$, lists $n = 51$. The entry is the Kaikkonen–Rosendahl code of 2003 [2], of density $1327/1024 \approx 1.29590$. The infinite family of [1] (Theorem 5.7) has lengths

$$n(r) = 26 \cdot 2^{r/2-4} - 1 = 2^m(51+1) - 1, \qquad m = r/2 - 5.$$

The 51 in this formula is the length-51 code; the whole family is QM_2^2 iterated from that single table entry. Theorem 5.7(i) of [1] calls the 2-partitions of the seed "very important for the next iterative process". Their computer search gives $p(\boldsymbol{H}_{KR}) = 11$ for the length-51 matrix.

A length-50 matrix at $r = 10$ therefore improves one table entry, and if it also admits a small $(2,0)$-partition, it re-seeds the whole family. The matrix of Theorem 1 and the partition of Table 2 do both.

2 Results

2.1 The code

Theorem 1. *Let $\boldsymbol{H}$ be the 10×50 matrix over $\mathbb{F}_2$ whose columns are the 50 integers of Table 1, where bit i of an integer, counted from the least significant bit, is row $i+1$. Then $\boldsymbol{H}$ has rank 10, its columns are distinct and nonzero, and condition (1) holds. The code with parity-check matrix $\boldsymbol{H}$ is a $[50, 40]_2$ code with covering radius exactly 2, and*

$$\ell_2(10, 2) \leq 50.$$

The proof is a finite computation over the committed matrix. Section 4 describes the two independent programs that perform it and the commands that replay it. Appendix A prints the matrix as bits.

1	2	4	15	16	32	65	86	128	173
183	202	212	247	256	297	320	329	341	366
373	381	391	403	438	460	479	491	502	559
576	608	653	734	742	754	771	777	789	821
846	855	869	881	893	897	927	981	1003	1004

Table 1: The columns of $\boldsymbol{H}$ as LSB-first integers.

The $1276 = 1 + 50 + \binom{50}{2}$ sums of at most two columns hit the 1024 syndromes with multiplicity histogram $\{1{:}859,\ 2{:}129,\ 4{:}24,\ 5{:}9,\ 6{:}3\}$. No multiplicity is zero, which proves (1). Exactly 973 syndromes are neither 0 nor a column, so the covering radius is not 1. The density is $\mu = 1276/1024 = 319/256 = 1.24609375$. The columns are distinct and nonzero, and $\boldsymbol{H}$ has a linearly dependent column triple, so the minimum distance is $d = 3$.

claim	value
shape	10×50, $\mathbb{F}_2$-rank 10, 50 distinct nonzero columns
coverage	1024/1024 syndromes
covering radius	exactly 2
density μ	$319/256 = 1.24609375$
minimum distance	$d = 3$
KR baseline	51 columns, 1024/1024, density 1327/1024

2.2 Minimality

Every matrix obtained from $\boldsymbol{H}$ by deleting one column violates (1): each single deletion leaves at least 9 of the 1024 syndromes uncovered, and the minimum, 9, is attained by exactly three columns, 381, 479, and 927. The column set of $\boldsymbol{H}$ is a minimal 1-saturating set in $PG(9, 2)$, and the covering-code literature calls such a code locally optimal. Minimality applies only to this particular 50-set; it does not decide whether a covering 49-set exists.

2.3 A $(2, 0)$-partition into ten blocks

The columns of $\boldsymbol{H}$ admit a $(2, 0)$-partition with $p(\boldsymbol{H}) = 10$, shown in Table 2.

block	columns
0	$2, 128, 202, 212, 771, 855, 897, 981$
1	86
2	$381, 893, 1003$
3	$183, 297$
4	$1, 65, 247, 256, 320, 438, 502, 734$
5	$15, 173, 329, 366, 460, 559, 653, 846$
6	$4, 16, 391, 491, 742, 754, 869, 881$
7	$479, 1004$
8	403
9	$32, 341, 373, 576, 608, 777, 789, 821, 927$

Table 2: A $(2, 0)$-partition of the columns of $\boldsymbol{H}$ with $p(\boldsymbol{H}) = 10$.

Each of the 973 syndromes that need a pair has a pair whose two columns lie in distinct blocks. Over those 973 syndromes the pair multiplicities are {1:821, 2:123, 4:19, 5:8, 6:2}. In particular, 821 syndromes have a unique pair, and each unique pair forces its two columns into distinct blocks. For comparison, the computer search of [1] gives $p(\boldsymbol{H}_{KR}) = 11$ for the length-51 code. Minimality of the 10-block partition is not claimed.

2.4 Dependent triples

$\boldsymbol{H}$ has exactly 10 linearly dependent triples of columns. Exactly one of them, $(491, 734, 821)$, meets three distinct blocks of the partition (blocks 6, 4, and 9). This is the analogue of Theorem 5.2(ii) of [1] and is the extra hypothesis that Construction QM_5^2 needs at $r = 28$. That construction is not carried out here.

2.5 Propagation

Construction QM_2^2 (Theorem 4.1 of [1]) takes an $[n_0, n_0 - r_0]_2\, 2$ code whose parity-check matrix $\boldsymbol{H}_0$ has a $(2,0)$-partition into $p(\boldsymbol{H}_0)$ blocks, together with an integer m such that $n_0 \geq 2^m \geq p(\boldsymbol{H}_0)$. In the notation of [1], its indicator set is $\mathscr{B} = \mathbb{F}_{2^m}$ and its tail block is $\boldsymbol{D} = \boldsymbol{D}_1(2)$. The output is a code with

$$n = 2^m(n_0 + 1) - 1, \qquad r = r_0 + 2m, \qquad p(\boldsymbol{H}_C) \leq 2^{m+1} + 1. \tag{2}$$

With $n_0 = 50$ and $p(\boldsymbol{H}_0) = 10$ the hypothesis permits exactly $m = 4$ and $m = 5$, since $2^6 = 64 > 50$. Both outputs come from equations (4.2) and (4.4) of [1] and were verified by enumeration of every syndrome:

	r	n	coverage	density	published
seed	10	50	1024/1024	319/256	51 (KR 2003)
$m = 4$	18	815	262144/262144	332521/262144	831
$m = 5$	20	1631	1048576/1048576	1330897/1048576	1663

The indicator assignment inside QM_2^2 is a free choice. A second assignment gives a second pair of matrices, committed under `result/data/alt/`, and both pairs pass the same verification.

2.6 The family

Iterating (2) from $(r, p) = (10, 10)$, using the partition bound $p(\boldsymbol{H}_C) \leq 2^{m+1} + 1$ at every step, reaches the even $r \leq 64$ accessible by QM_2^2:

$$10, 18, 20, 30, 32, 34, 36, 38, 40, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64.$$

The even r not reachable this way are $12, 14, 16, 22, 24, 26, 28, 42, 44$. The paper fills its own gaps at $r = 22, 24, 26$ with QM_3^2 and at $r = 28$ with QM_5^2. On the reachable r the lengths and densities have the closed form

$$n = 51 \cdot 2^{r/2-5} - 1, \qquad \mu = \frac{51^2}{2^{11}} - \frac{51}{2^{r/2+1}} + \frac{1}{2^r} \nearrow \frac{2601}{2048} \approx 1.27002.$$

Hence $\bar{\mu}(2) \leq 2601/2048$, and therefore $f(2) \leq 2601/2048$.

	$\bar{\mu}(2)$	source
pre-2025	$729/512 \approx 1.42383$	standing since 1992
arXiv:2511.02542	$2704/2048 \approx 1.32031$	seeded by KR $n_0 = 51$
this seed	$2601/2048 \approx 1.27002$	seeded by $n_0 = 50$

The family rows past $m = 5$ use the bound $p(\boldsymbol{H}_C) \leq 2^{m+1}+1$ of [1], not computed partitions.

3 The search

3.1 The matrix

The 50-column matrix $\boldsymbol{H}$ was found by simulated annealing on column sets in $\mathbb{F}_2^{10}$. A state is a set of exactly 50 distinct nonzero columns. An array of 1024 counters records, for each syndrome, its number of representations among the $1 + 50 + \binom{50}{2} = 1276$ sums of at most two columns. The energy of a state is its number of zero counters. Replacing one column changes only that singleton and the 49 pair-sums that use it, so a swap costs about 100 counter updates instead of a 1276-term recount. Most proposals target an uncovered syndrome g: the search offers g itself, or $g + s$ for a current column s.

The published $r = 10$ seed is the Kaikkonen–Rosendahl length-51 matrix. Before any partition work it was reconstructed from the hexadecimal listing in Theorem 4.3 of [1]. That listing is MSB-first; the repository stores columns LSB-first (Section 4.3). The reconstructed matrix has 51 distinct columns and covers 1024/1024 syndromes, so it is the published table entry. The one-column improvement below is an improvement of that seed, not a different problem.

The anneal starts from the best single deletion of this 51-set: deleting column 419 (`0x1A3`) leaves 11 syndromes uncovered. Energy is the number of uncovered syndromes. Moves that do not raise energy always pass; moves that raise it by Δ pass with probability $e^{-\Delta/T}$ (Metropolis). With a fixed xorshift64 state, the first run of the script reached a covering 50-set at proposal 3,600,281. The search stopped there. It did not compute a $(2, 0)$-partition and it did not run Construction QM_2^2.

An independent check of the resulting matrix, from the committed column text, gives a 10×50 matrix of $\mathbb{F}_2$-rank 10 with 50 distinct nonzero columns, coverage 1024/1024, and at least one syndrome that is neither 0 nor a column. That finite check is the one-column improvement of Table 5.1 of [1].

The same method failed at three nearby parameters. At $(r, n) = (8, 25)$ the best anneal left 3 of 256 syndromes uncovered. At $(9, 38)$ it left 8 of 512. At $(10, 49)$ it left 7 of 1024. Those are incomplete searches, not lower bounds.

3.2 The partition

The published $r = 10$ row is the seed of the whole QM_2^2 family (Theorem 5.7 of [1]). A covering 50-set is enough for $\ell_2(10, 2) \leq 50$. The family bound needs a $(2, 0)$-partition with few enough blocks. That was a second computation.

Of the 1024 syndromes, 973 need a pair. Their pair-multiplicities are

$$\{1{:}821,\ 2{:}123,\ 4{:}19,\ 5{:}8,\ 6{:}2\}.$$

Each of the 821 unique-pair syndromes forces its two columns into distinct blocks. These constraints form a graph with 821 edges on the 50 columns; the maximum degree is 42. A valid $(2, 0)$-partition is a proper colouring of this graph that, for each of the remaining 152 syndromes, also splits at least one of its 2 to 6 pairs. Checking a candidate is one pass over the 1276 sums of at most two columns.

A colouring with 10 blocks exists and verifies; it is the artifact `result/data/partition_p10.json`. The paper takes $p(\boldsymbol{H}) = 10$ from that file, not from a later rerun of the search. The same search found colourings with 16, 12, and 11 blocks on the way down, and did not find a 9-block or 8-block colouring in that run. Not finding nine blocks is an incomplete search, not a lower bound on $p(\boldsymbol{H})$. Ten blocks is smaller than the computer-searched $p(\boldsymbol{H}_{KR}) = 11$ of [1], and small enough that $n_0 \geq 2^m \geq p(\boldsymbol{H}_0)$ holds for both $m = 4$ and $m = 5$.

The discovery programs that certified the 50-set, reconstructed the Kaikkonen–Rosendahl seed, searched for the partition, and enumerated the $m = 4, 5$ outputs sit in `problems/covering/compute/`. They are not the two independent verifiers of Section 4.

3.3 The check

Construction QM_2^2 was implemented from (4.2) and (4.4) of [1]. The $m = 4$ and $m = 5$ outputs were enumerated over their full syndrome spaces: length 815 at $r = 18$ covers 262144/262144 syndromes, density $\mu = 1.26847$, against the published length 831; length 1631 at $r = 20$ covers its full space, against the published length 1663. The two opposite verifiers of Section 4 re-derive the columns from the matrix text and use the partition file for block labels only. After they agreed, a third program outside the repository was run once. The covering radius of $\boldsymbol{H}$ is that finite check, not a property of the annealing path.

4 Verification

4.1 Replay

Clone the public repository, check out commit `82b5b85bd14bfa42ac7439adc7c7c956c3f84eeb`, and run the pipeline from this pin:

```
git clone https://github.com/wustep/maths.git
cd maths
git checkout 82b5b85bd14bfa42ac7439adc7c7c956c3f84eeb
cd problems/covering/share/2026-08-24/result
./run_all.sh
```

The pipeline needs `python3` and `rustc`. After the checkout it uses no network, no packages, and no randomness. It executes 74 named assertions, exits with status 0, and prints byte-identical output across runs. A run takes about two seconds. Direct links at that commit: `https://github.com/wustep/maths/tree/82b5b85bd14bfa42ac7439adc7c7c956c3f84eeb/problems/covering/share/2026-08-24/result` and `result/run_all.sh`.

4.2 Two verifiers with opposite algorithms

`verify/verify.py` is pair-driven: it enumerates all $\binom{n}{2}$ pairs, marks the syndromes they reach, then sweeps all 2^r values for unmarked ones. `verify/verify.rs` is syndrome-driven: for each of the 2^r syndromes s it scans the columns h and tests membership of $s \oplus h$. It then runs its own pair loop and reports only if its two verdicts agree. The two programs also differ in parser and in rank pivot choice, lowest versus highest set bit, and both use exact rational arithmetic. `run_all.sh` compares their fact dumps line by line.

Both programs read the matrix from text. Neither reads a stored certificate. After both programs agreed, a third program, kept outside the repository, was run once as a post-hoc comparison and confirmed every value; it is not in the tree and is not part of `./run_all.sh`.

4.3 Encoding

Every file in the repository is LSB-first: bit i of a column integer is row $i+1$. The hex listing of M_{KR} in Theorem 4.3 of [1] is MSB-first, with row 1 as the high bit. Reconstruction of $\boldsymbol{H}_{KR} = [\, I_{10} \mid M_{KR} \,]$ therefore reverses the ten bits of each column. A verifier that misses the reversal fails on the Kaikkonen–Rosendahl baseline and only there, which makes the mistake visible. The builder asserts the relation $h_5 + h_{27} + h_{29} = 0$ of Theorem 5.2(ii) of [1] as a guard.

A reader who uses the MSB-first convention obtains the bit-reversal of every column. Bit reversal is an invertible linear map on $\mathbb{F}_2^{10}$, so every claim in this note is preserved.

5 What is not claimed

- *Optimality.* The sphere-covering bound gives $\ell_2(10,2) \geq 45$, since $1 + n + \binom{n}{2} \geq 1024$ fails at $n = 44$ with 991. Minimality concerns this 50-set only. A covering 49-set may exist.
- *Nearby parameters.* The searches at $(10,49)$, $(9,38)$, and $(8,25)$, which left 7, 8, and 3 syndromes uncovered respectively, are incomplete searches, not lower bounds.

- *Exhaustive checks.* Only the $m = 4$ and $m = 5$ outputs are verified exhaustively. The later rows of the family inherit $p(\boldsymbol{H}_C) \le 2^{m+1} + 1$ from (4.4) of [1].
- *Gaps.* Nothing is claimed here for $r \in \{12, 14, 16, 22, 24, 26, 28, 42, 44\}$.
- *Asymptotics.* $f(2) \le 2601/2048$ is an upper bound only. Whether $f(2) = 1$ is open.
- *Priority.* Table 5.1 of [1] is "best as far as the authors know". A proof of $\ell_2(10, 2) \le 50$ may exist in older proceedings.

6 Files

Everything named below is in the public repository `https://github.com/wustep/maths`, under `problems/covering/share/2026-08-24/` at commit `82b5b85bd14bfa42ac7439adc7c7c956c3f84eeb`.

path	contents
`result/`	the verification pipeline and matrices
`result/run_all.sh`	the replay script
`result/data/H_r10_n50.txt`	the matrix
`result/data/partition_p10.json`	the 10-block partition
`compute/`	the annealing replay
`problems/covering/compute/`	discovery programs (live tree)

A The matrix as bits

Rows 1 to 10 from top to bottom, columns left to right. Bit i (LSB) of each integer column is row $i + 1$.

```
10010010011001010110111100110100100011110111111110
01010001001101000001001110111100011110001100001010
00110001011011000011111011101100111000111110101101
00010000010100010101010001110100110001001000101011
00001001001011000010110110101000010100110101101100
00000100011001010001110010011101001100010011100011
00000011000111001111110001111011011100001111100111
00000000111111000000001111111000111100000000011111
00000000000000111111111111111000000011111111111111
00000000000000000000000000000111111111111111111111
```

References

[1] A. A. Davydov, S. Marcugini, F. Pambianco, *New upper bounds for binary linear covering codes*, arXiv:2511.02542 [cs.IT], November 2025.

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[3] G. Cohen, I. Honkala, S. Litsyn, A. Lobstein, *Covering Codes*, North-Holland Mathematical Library **54**, Elsevier, Amsterdam, 1997.

[4] B. Green, *100 open problems*, Problem 40. `https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf`.